

PRODUCT REQUIREMENTS DOCUMENT

Aura Pulse 90

An AI-Native Fitness Consistency System

Arunjyoti Kalita

Product Owner · v1.0 — Shipped May 2026

aura-pulse90.web.app

Field	Value
Product	Aura Pulse 90
Version	1.0 — MVP (Shipped)
Status	Live · aura-pulse90.web.app
Platform	Progressive Web App — Android-first, cross-browser
Product Owner	Arunjyoti Kalita
Last Updated	May 2026

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Why this product exists

Fitness app churn is a product architecture problem, not a motivation problem. Existing apps optimise for activity logging — they capture behaviour without sustaining it. Three structural failures drive this:

Failure Mode	Mechanism	Design Response
Paywalled coaching	Habit loops interrupted at 7–14 days — exactly when routines are forming	Full AI coaching free from day one; zero freemium gate
Open-ended structure	No finish line creates sustained motivational load; users disengage when one day breaks	Bounded 90-day arc with re-engagement checkpoints at Days 14, 28, 56, 84
Western food defaults	Daily friction for South Asian users forces manual meal configuration at every session	South Asian defaults loaded at onboarding; no configuration required
Generic AI advice	Context-free coaching erodes trust — users disengage when advice ignores their actual state	15-signal context pipeline ensures coaching reflects real-time user state
Shallow progression	Streak-only systems reward presence, not progress — no identity reinforcement over time	XP + 14 badges + 6 level titles tied to transformation milestones
Missed day = dead end	Most apps show nothing when a workout is skipped — churn risk peaks at exactly this moment	Skip flow immediately surfaces Quick Sessions (5/10/15 min + Desk Mode) — every skip becomes a re-engagement moment

Product vision

A bounded, AI-coached, culturally contextual fitness system — free, forever — achieving stronger behavioral retention than subscription apps by eliminating cost, cultural, and structural friction simultaneously.

Out of scope — v1.0

Explicitly deferred to maintain MVP focus and execution velocity:

- Native iOS / Android apps — PWA distribution only
- Social features beyond leaderboard — comments, DMs, group challenges
- Paid subscription tier — all features free in v1.0
- Third-party wearable integrations — Apple Watch, Fitbit, Garmin
- Nutritionist-reviewed meal plans — requires specialist involvement at scale

Primary persona — The Restarter

Attribute	Profile
Age	18–35
Location	Tier 2/3 Indian city — Guwahati, Indore, Patna, Coimbatore
Background	Student or early-career professional
Device	Android, limited data plan
Diet	South Asian — dal, roti, rice, paneer, oats, soya chunks
App history	Tried 2–4 apps; quit when trial expired or app felt irrelevant
Core frustration	₹1500–2,000/mo pricing and Western food suggestions
Core motivation	Structured transformation with visible, measurable progress

Secondary persona — The Experimenter

Attribute	Profile
Age	22–45
Location	Metro city or South Asian diaspora (Mumbai, Delhi)
Motivation	AI accountability without personal trainer costs
Behaviour	Heavy AI coach usage; lower diet module engagement

User needs — ranked by observed signal strength

Need	Signal	Product Response
Culturally relevant diet	Very High — most-discussed feature, all sessions, unprompted	South Asian defaults from day one; zero configuration
Bounded program structure	High — 'no clear goal' cited as primary churn reason	90-day arc + milestone checkpoints at Days 14, 28, 56, 84
Free full-feature access	High — paywall friction cited unprompted by all early users	100% free; no freemium gate post-signup
Daily accountability loop	High — daily return without push prompts observed	Streak + XP + dashboard + AI morning guidance
Contextual AI coaching	Medium — emerged after users experienced generic AI elsewhere	15-signal context pipeline per coaching interaction
Recovery awareness	Medium — surfaced in Week 3 feedback	Composite recovery score: sleep + stress + intensity
Safe solo training guidance	Medium — users training without supervision	TF.js MoveNet form check with real-time rep counting

What v1.0 was designed to validate

Validation goals — MVP

- 1. **G1** — Behavioral retention loop — do users return daily without push notification prompts?
- 2. **G2** — Cultural positioning — does the South Asian diet module reduce activation friction?
- 3. **G3** — AI coaching quality — do users find Coach Max responses contextually relevant?
- 4. **G4** — Execution feasibility — can a full-stack AI PWA ship in under 4 weeks, PM-led, no eng team?

MVP signal — observed vs. targeted

v1.0 shipped without analytics instrumentation. Signals are behavioral, observed across live user sessions.

Metric	Target	Signal Observed	Verdict
Daily active return	D7 >50%	Users returned daily without prompts across all sessions	✓ Validated
Diet module reception	Positive feedback	Most-discussed feature — unprompted, consistent across users	✓ Strongly validated
AI coach relevance	>80% contextual	Users referenced specific coach responses in follow-up sessions	✓ Validated
Onboarding completion	>70%	No analytics — unmeasured	⚠ Gap
Build velocity	<6 weeks	Shipped in 4 weeks, part-time	✓ Exceeded
Streak persistence	Avg >3 days	Multi-day streaks maintained across multiple sessions	✓ Validated

Phase 2 targets — requires analytics first

None of these targets can be measured until PostHog instrumentation ships in Phase 2.

Metric	P2 Target	Rationale	Tool
D7 Retention	≥40%	Primary signal for behavioral loop quality	PostHog
D30 Retention	≥20%	Validates 90-day program engagement	PostHog
90-Day Completion	≥15%	Core product hypothesis — bounded programs	PostHog
First Workout Conv.	≥65%	Measures onboarding friction	PostHog
Coach Sessions/Wk	≥3 per DAU	Validates AI coaching as retention driver	Firebase logs
Streak Recovery	≥30% in 3d	Measures re-engagement system effectiveness	PostHog

04 — USER STORIES

Feature requirements

P0 = Must-have for launch. P1 = Strong preference. P2 = Nice-to-have. All stories shipped in v1.0 unless noted.

Authentication & Onboarding

ID	As a...	I want to...	So that...	Priority
AUTH-01	new user	sign in via Google in one tap	I can start without creating a new password	P0
AUTH-02	new user	sign in via email and password	I have an alternative to Google sign-in	P0
AUTH-03	new user	see program context after sign-in	I understand what the 90-day arc involves before starting	P1
AUTH-04	returning user	data to persist across sessions and devices	streak, XP, and history survive a phone switch	P0

Workout & Core Loop

ID	As a...	I want to...	So that...	Priority
WRK-01	user	log a workout with exercise, sets, reps, weight	I have a complete training history	P0
WRK-02	user	see today's recommended workout on the dashboard	I don't have to decide what to do — I just start	P0
WRK-03	user	earn XP every time I complete a workout	progress is visible beyond the workout log itself	P1
WRK-04	user	see workout history and strength progression	I can verify I'm actually getting stronger over time	P1
WRK-05	user	use AI form checking during exercises	I can train safely without a trainer present	P2
WRK-06	user who skipped today	see quick micro-session options immediately after skipping	I can keep momentum without completing the full planned session	P1

Diet & Nutrition

ID	As a...	I want to...	So that...	Priority
DIET-01	South Asian user	familiar foods as default meal options	dal and paneer are there from day one — no search required	P0

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DIET-02	user	log meals and see macro breakdown	I understand whether I'm hitting protein and calorie targets	P0
DIET-03	user	a default South Asian meal plan each day	I have a starting point, not a blank slate	P1
DIET-04	user	track daily water intake	hydration is logged alongside food	P2
DIET-05	user	scan food barcodes for fast logging	meal logging takes seconds	P2

AI Coach — Coach Max

ID	As a...	I want to...	So that...	Priority
AI-01	user	coaching guidance that reflects my actual state today	advice feels personal, not generic	P0
AI-02	user	ask the coach questions about workouts, diet, and recovery	I have an always-available expert who knows my history	P0
AI-03	user	the coach to reference my streak, XP, and recovery score	responses reflect my actual progress, not a template	P1
AI-04	user	a useful fallback message when the AI call fails	a network error doesn't break my experience	P1

Recovery & Wellbeing

ID	As a...	I want to...	So that...	Priority
REC-01	user	log sleep quality and hours each night	the system has accurate recovery input	P1
REC-02	user	log daily stress level	recovery score reflects mental as well as physical state	P1
REC-03	user	see a composite recovery score each morning	I can make an informed intensity decision for the day	P1
REC-04	user	AI coaching adjusted by my recovery score	the coach knows when to push and when to advise rest	P1
REC-05	user	guided breathing exercises from within the app	stress management earns XP and integrates into the program	P2

Gamification & Progression

ID	As a...	I want to...	So that...	Priority
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GAM-01	user	earn XP for every positive health action	every contribution — not just workouts — feels rewarded	P1
GAM-02	user	see streak and level on the dashboard	consistency is visible and motivating at a glance	P1
GAM-03	user	earn badges for transformation milestones	I have visible evidence of how far I've come	P2
GAM-04	user	a community leaderboard of XP	I have social accountability alongside personal motivation	P2

05 — ARCHITECTURE DECISIONS

Five decisions that defined the product

Decision 1 — Local-First Data Architecture

PROBLEM	Daily-use app requiring instant load for users in low-connectivity Tier 2/3 markets, with zero tolerance for per-request cloud costs at free tier.
DECISION	localStorage as primary state store + Firebase as async cloud backup, synced via debounced event bus (1.5s delay).
WHY	Sub-second load times with no network dependency for daily interactions. Offline-first behavior for unreliable connectivity. Zero Firestore read costs at MVP scale — keeps the product permanently free.
TRADEOFF	5MB localStorage ceiling limits power users with large photo logs. Accepted as a known Phase 2 migration trigger — Firebase Storage upgrade activated by usage scale, not day-one infrastructure spend.

Decision 2 — Authentication Required, No Guest Mode

PROBLEM	AI coaching personalisation and cross-device data continuity require persistent user identity. Leaderboard integrity depends on authenticated participation.
DECISION	Sign-in required at onboarding: Google OAuth or Email/Password. No anonymous access path.
WHY	The product's core value — AI coaching calibrated to weeks of behavioral history — cannot be delivered without identity. A guest flow would produce a materially weaker product and create a misleading first impression.
TRADEOFF	Signup friction at activation. Accepted because identity gates the highest-value features. Users who complete sign-in receive a product that earns the additional step.

Decision 3 — South Asian-First Diet Defaults

PROBLEM	Global fitness apps default to Western food databases — creating daily manual configuration friction for South Asian users that compounds into abandonment.
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DECISION	South Asian meal defaults loaded at first launch — no configuration required. Oats + peanut butter breakfast, dal + brown rice lunch, paneer/soya dinner.
WHY	Eliminates the highest daily friction point for the primary persona. Signals the product as built for this market — not adapted for it as an afterthought. Creates a defensible differentiator that requires cultural knowledge, not just engineering resources.
TRADEOFF	Narrower initial TAM. Expansion to diaspora and metro users with mixed eating patterns requires a broader food database in Phase 2.

Decision 4 — Dynamic CDN Loading for TF.js Form Check

PROBLEM	AI Form Check requires a 4MB TensorFlow.js model. Bundling it degrades initial load for all users, 90% of whom never use the feature.
DECISION	Dynamic CDN import of MoveNet model triggered only on Form Check open. Model never downloaded for users who don't need it.
WHY	Preserves sub-second initial load across the full user base. Form Check is a high-value differentiator that justifies a CDN dependency for the minority who use it.
TRADEOFF	Creates a fragile loading path specifically for Form Check. Mitigated by graceful degradation to manual rep counting if model load fails.

Decision 5 — 60+ Feature Flags as Experimentation Architecture

PROBLEM	MVP scope must be controlled without removing features from the codebase. Future premium tier requires activation without a new release cycle.
DECISION	Every feature sits behind a boolean flag — toggled independently per environment or per user group, without deployment.
WHY	Instant rollback of any feature. A/B testing scaffolding from day one. Future premium tier = enable flags for paying users — zero structural change to the product.
TRADEOFF	Flag management complexity grows at scale. Requires a flag registry and audit process before significant user base expansion.

Coach Max — context architecture

AI coaching quality is determined entirely by context architecture — not by the model. The 15-signal pipeline below defines what Coach Max knows about a user before generating any response.

Signal	Data Source	Coaching Impact
Recovery score (0–100)	Sleep + stress + intensity algo	Determines intensity recommendation — push vs. rest
Current streak (days)	localStorage / Firestore	Calibrates motivational tone — encouragement vs. challenge
Program day (1–90)	Program state	Positions advice within the transformation arc
Last workout type	Workout log	Avoids same-muscle repetition; suggests complementary session
Last workout intensity	Workout log	Direct input to recovery score and volume recommendations
Sleep hours (last night)	Sleep log	Low sleep → conservative intensity guidance
Sleep quality (1–5)	Sleep log	Subjective energy signal for cognitive and physical load
Stress level (1–5)	Daily check-in	High stress → lighter session, recovery prioritised
Mood check-in	Daily check-in	Shapes coaching tone — empathetic vs. motivating
Current XP total	Gamification state	Milestone proximity creates coaching hooks
Level / title	Gamification state	Reinforces identity and transformation narrative
Recent badges earned	Gamification state	Badge milestones are coaching engagement opportunities
Macro compliance (today)	Diet log	Nutrition state informs energy and performance advice
Calorie balance	Diet log	Deficit / surplus shapes workout and recovery recommendations

AI Form Check — technical specification

Component	Specification
Model	TensorFlow.js MoveNet SinglePose Lightning
Load strategy	Dynamic CDN import — not bundled — removes ~4MB from initial app load
Detection	17 body keypoints at 30fps on mid-range Android hardware
Rep counting	Joint angle detection with configurable threshold per exercise
Voice feedback	Web Speech API — rep count and form deviation alerts
Exercises (v1.0)	Squat, push-up, shoulder press, bicep curl, deadlift
Failure behaviour	Graceful degradation to manual rep counting if model load fails

AI failure and degradation requirements

Non-negotiable failure mode design

- Gemini API failure → surface a static, contextually appropriate message. Never show 'Something went wrong.' Coach must remain present even when AI is unavailable.
- TF.js CDN load failure → gracefully degrade to manual rep counting. Inform user without blocking the workout session.
- Firebase sync failure → continue from localStorage. Queue sync for next successful connection. Never block user actions on sync state.
- All AI-dependent features must function in a degraded but usable state without network connectivity.

07 — FEATURE FLAG REGISTRY

Selected flag register — highest priority

All features are flag-controlled. Default state reflects v1.0 launch configuration.

Flag	Default	Purpose & Trigger
ENABLE_AI_COACH	ON	Master coach toggle — disable if Gemini quota exceeded or cost risk detected
ENABLE_FORM_CHECK	ON	TF.js MoveNet — disable on low-end devices or CDN failure
ENABLE_LEADERBOARD	ON	Community XP board — disable if abuse or gaming detected
ENABLE_BREATHING_MODULE	ON	Guided breathing with XP reward — mental wellness integration

ENABLE_SKIP_QUICK_SESSION	ON	Post-skip micro-workout offer (5/10/15 min + Desk Mode) — disable to A/B test re-engagement approach
ENABLE_BARCODE_SCANNER	OFF	Food barcode scan — Phase 2 feature; localStorage impact
ENABLE_MEAL_PHOTOS	OFF	Photo meal logging — Phase 2; localStorage ceiling risk
ENABLE_SOCIAL_SHARING	OFF	Badge + milestone card sharing — Phase 2 viral feature
ENABLE_PREMIUM_FEATURES	OFF	Future premium tier scaffold — activates without code change
ENABLE_EXERCISE_SNACKS	OFF	5-min workouts for busy days — Phase 2 retention feature
ENABLE_PROGRESS_PHOTOS	OFF	Before/after photo storage — Phase 2; storage migration required first
ENABLE_COACH_STREAMING	OFF	Streaming AI responses — requires Gemini 2.0 upgrade
ENABLE_ANALYTICS	OFF	PostHog integration — Phase 2 P0; blocks data-driven iteration
ENABLE_PUSH_NOTIFICATIONS	OFF	Streak reminders via web push — Phase 2 retention feature

08 — RISK REGISTER

Known issues with mitigation plan

Issues are prioritised by potential user impact and cost exposure, not by ease of fix.

Issue	Severity	User / Business Impact	Mitigation
Auth guard disabled on coachAction — anonymous callers can trigger Gemini API at owner cost	P0 — Critical	Unbounded API cost exposure before scaling	Re-enable Firebase auth check. Ship before any public launch push.
No analytics instrumentation	P1 — High	All product decisions rely on qualitative signal only — blocks data-driven iteration	PostHog day one of Phase 2. No Phase 3 decisions without funnel data.
localStorage 5MB ceiling	P1 — High	Power users with photo logs will lose data silently on overflow	Migrate to Firebase Storage in Phase 2. Current MVP users unlikely to hit limit.
Dashboard.tsx monolith (~2000 lines)	P1 — High	Slows feature velocity; high risk of regression on any change	Decompose into sub-components in Phase 2 before team scales.
No per-page error boundaries	P2 — Med	Single component crash takes down entire app	Add React error boundaries per route in Phase 1 stability sprint.
Supabase legacy config in codebase	P2 — Med	Dead dependency and minor security hygiene risk	Remove in Phase 1 cleanup. One-line fix.

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No unit tests for core algorithms	P2 — Med	Recovery score, streak, diet score bugs at scale are undetectable	Add unit test coverage in Phase 2 before expanding user base.
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09 — ROADMAP

Phase-gated delivery plan

Phase gates are trigger-based, not purely calendar-based. Phase 2 does not begin until Phase 1 security items are resolved.

Phase 1 — Stability (Weeks 1–2)	Phase 2 — Scale (Month 2)	Phase 3 — Growth (Month 3+)
Auth guard on coachAction — P0 cost risk	PostHog analytics — blocks all data-driven decisions	Gemini 2.0 with streaming coach responses
Migrate photos → Firebase Storage — removes 5MB ceiling	Migrate primary state → Firestore	Native mobile via Capacitor — App Store + Play Store
Remove Supabase legacy config — security hygiene	Decompose Dashboard + WorkoutPage monoliths	Social milestone sharing — organic acquisition
Add per-page error boundaries	Unit tests: recovery score, streak, diet score algos	Premium tier — ₹299–₹799/mo advanced AI features
	Re-engagement flow design — addresses highest churn risk	B2B white-label — ₹5,000–₹20,000/mo per gym/coach
	ENABLE_ANALYTICS flag → ON	Expanded South Asian + diaspora food database

10 — OPEN QUESTIONS

Decisions deferred — with decision triggers

These questions were intentionally left open in v1.0. Each has a clear trigger condition before a decision can be made responsibly.

Question	Why It's Open	Decision Trigger
What is the right premium price point?	No conversion data exists to test price sensitivity	A/B test at ₹99, ₹299, ₹499, ₹799 — requires PostHog first
Notification vs. in-app re-engagement?	Opt-in rate unknown until Phase 2 web push ships	Test both variants against D7 and D30 retention impact
B2B: gym-first or coach-first?	No discovery interviews completed with either segment	5–10 interviews with gym owners and independent trainers
Generic vs. specialised 90-day programs?	Current structure is generalised — fat loss, muscle, endurance variants may improve completion	Requires 90-day completion rate data from Phase 2

Global vs. friend-circle leaderboard?	Global board may demotivate beginners — unvalidated	A/B test with Phase 2 user cohort; measure streak impact
When does diet module need dietitian review?	Built on general nutritional knowledge; adequate at MVP scale	Required before public health claims or >10,000 active users
Which markets to expand to post-India?	South Asian diaspora is logical but unvalidated as a segment	Instrument diaspora registration rate from Phase 2 analytics

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